SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	**ITA05**	**Course Title:**	**Computer Vision**
CO Assessed:	**CO4 – Articulation (combining methods for evaluation) P4**	**Assessment:**	**Assessment Tool 2 – Industry Problem Solving Task**
Weightage:	**20% of CIA**	**Total Marks:**	**100**
CO4 Statement:	*Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)*		
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Section / Batch:	Third year/ B.Tech IT	**Date:**	19/08/2026

Code of Conduct

I __Kaviya V/192421022___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___Kaviya V______

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

QUESTION: Retail Motion Tracking Logic

Scenario:

A retail store uses a Computer Vision system to track customer movement. The system must work in two different conditions:

- Low crowd density: Accurate tracking of individual customers is required.
- High crowd density: Tracking should remain stable even when customers overlap or move close together.

The given observations are:

Method	Low Crowd Density	High Crowd Density
Optical flow	Accurate	Noisy
Motion models	Less detailed	Stable

This is a trade-off between tracking accuracy/detail and tracking stability, which is important when designing real-time Computer Vision systems.

Understanding of the Problem:

The main challenge is that one tracking technique does not perform equally well in both crowd conditions.

- In a low-density crowd, customers are separated clearly, so individual motion can be estimated accurately.
- In a high-density crowd, people may overlap, occlude one another, and move in different directions.
- Optical Flow can provide detailed motion information but becomes noisy when many people are close together.
- Motion Models provide more stable tracking in crowded situations but may lose detailed individual motion information.

Therefore, the system needs an approach that can adapt to changing crowd density.

Comparison of the Two Approaches:

Optical Flow:
Advantages:

- Provides detailed pixel-level motion information.
- Works accurately when objects are clearly separated.
- Suitable for tracking individual customer movement.
- Useful when crowd density is low.

Limitations:

- Can become noisy when many people are present.
- Occlusion and overlapping customers can cause incorrect motion vectors.
- Tracking becomes less reliable in highly crowded areas.

Motion Models:
Advantages:
- Provides stable estimation of object movement.
- More suitable for crowded environments.
- Can maintain tracking when individual motion information becomes unreliable.
- Less affected by noisy optical-flow measurements.

Limitations:
- Provides less detailed motion information.
- May not accurately represent sudden or complex individual movements.
- Less suitable when precise individual tracking is required.

Application of Computer Vision Concepts:

A suitable solution is to use a hybrid/adaptive tracking approach.

Proposed Workflow:

Camera Input
↓
Customer Detection
↓
Estimate Crowd Density
↓
Low Density? → Optical Flow
↓
High Density? → Motion Model
↓
Combine/Update Tracking Results
↓
Stable Customer Tracking

This approach uses the strengths of both techniques rather than depending on only one method.

Proposed Solution:

When Crowd Density is Low
Use Optical Flow because customers are separated and their movement can be accurately estimated.
For example:
Customer A moves from the entrance toward the clothing section → Optical Flow accurately estimates the direction and speed of movement.

When Crowd Density is High
Switch to or give more importance to the Motion Model.
For example:
Several customers are walking close together → Optical Flow produces noisy motion vectors, so the motion model provides a more stable estimate of their movement.

Adaptive Combination

The system can monitor crowd density continuously:

- Low density → Higher weight for Optical Flow
- High density → Higher weight for Motion Model
- Intermediate density → Combine both

This makes the system more robust to changing retail conditions.

Justification:

The hybrid approach is better than selecting only one technique.

Requirement	Optical Flow	Motion Model	Hybrid Approach
Low-density accuracy	High	Moderate	High
High-density stability	Low	High	High
Detailed motion	High	Moderate	High
Handling crowding	Low	Good	Good
Adaptability	Moderate	Moderate	High

Why Hybrid is Better

- If only Optical Flow is used, the system may lose tracking stability in crowded areas.
- If only Motion Models are used, the system may not capture detailed individual movement when the crowd is sparse.
- Therefore, combining both methods according to crowd density provides a better balance between accuracy, stability, and detail.

Final Conclusion:

The most suitable solution for the retail system is an adaptive hybrid tracking approach combining Optical Flow and Motion Models.

- Optical Flow should be preferred when crowd density is low because it provides accurate and detailed motion information.
- Motion Models should be preferred when crowd density is high because they provide more stable tracking.
- An adaptive system can switch or adjust the contribution of each method based on crowd density.

Final Answer:

Use Optical Flow for low-density crowds and Motion Models for high-density crowds. An adaptive hybrid approach is the most suitable because it combines the accuracy of Optical Flow with the stability of Motion Models, making the retail tracking system more robust under varying crowd conditions.